

Aditya Manoj

Developer I – Full-Stack & DevOps Engineering

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Professional Profile

Dynamic Computer Science Engineer with **2+ years** of experience spanning production full-stack engineering, zero-knowledge security architectures, interactive 3D graphics for modern landing pages, agentic AI prompt engineering, and CI/CD pipeline automation. Proven track record architecting offline-first Progressive Web Applications, high-concurrency event operating systems processing **1.2M+ operations** with 100% uptime, asset loading optimization for consumer platforms. **Meta-Certified Full Stack Developer** skilled in Python automation, Docker, WebGL/Canvas, Model Context Protocols (MCPs), and cloud platforms (Firebase, Vite, Edge CDN). Seeking to leverage cross-functional expertise in software delivery and infrastructure optimization to build **high-impact systems**.

DevOps & Technical Skills

- **DevOps & Cloud Infrastructure:** Python (OOP) — Git/GitHub Workflows — CI/CD Pipelines — Bash Scripting — Docker Basics — Firebase Hosting & Functions — Vite Edge Deployment
- **Frontend & Web Systems:** React / React Native — JavaScript (ES Modules) — IndexedDB (Dexie.js) — HTML5 Canvas / WebGL — Spline 3D — Framer Motion — CSS Grid / Tailwind
- **AI Systems & Full-Stack:** Agentic Specs / Prompt-Ops — Model Context Protocol (MCP) — Node.js / Express — FastAPI / Django — Firebase Firestore — WebAuthn / Passkeys

Certifications & Specializations

Meta Full Stack Developer Specialization

2024–2025

Coursera (Meta)

- Full-stack mastery covering React, Python/Django, REST APIs, database design, HTML5/CSS3 and **production deployment**.

Professional Experience

Full-Stack PWA & Security Systems Engineer

Jan 2026 – Present

SBAFA Financial Enclave (sbafa-ft.web.app)

Remote

- Architected an **offline-first, zero-knowledge** Progressive Web Application (PWA) utilizing vanilla ES modules, IndexedDB (Dexie.js), and Firebase Firestore, orchestrating bi-directional reactive synchronization for 1,000+ ledger records.
- Engineered an on-device, sandboxed PDF statement ingestion engine leveraging pdfjs-dist and deterministic cryptographic passkey decryption, executing multi-bank **OCR parsing** with zero third-party telemetry.
- Implemented an automated OAuth2 Gmail pipeline to ingest and decrypt scheduled banking e-statements, deploying contextual regex heuristics for **zero-latency** transaction deduplication.
- Hardened client security and session durability via **WebAuthn passkeys**, SHA-256 PIN hashing, and multi-tier Service Worker caching strategies (Stale-While-Revalidate) to sustain offline financial tracking.

Lead PWA & Frontend Systems Engineer

Jul 2026 – Sep 2026

IndZita Biotech (indzita biotech.com)

Remote

- Architected an **offline-first** Progressive Web Application (PWA) leveraging custom Service Worker strategies and IndexedDB caching to serve a high-performance **3D visualization suite** (Catmull-Rom splines, Euler kinematics) with zero load-time overhead.
- Built an installable, mobile-optimized **PWA app shell** and reactive genomic canvas, optimizing frame-budget lifecycles via IntersectionObserver to sustain **fluid 60 FPS performance** across low-spec handheld devices.
- Engineered an offline-capable Virtual **Diagnostics Lab simulator** modeling real-time thermal PCR kinetics (C_t thresholds) and piRNA biomarker workflows, translating high-stakes requirements for clinical partners (**RGCB, Karkinos Healthcare, NIMHANS**) under the MAHA MedTech Mission.
- Provisioned PWA web app manifests, background sync handlers, and automated Vite edge deployment pipelines on Firebase Hosting, achieving **sub-second global delivery** and **zero-downtime reliability**.

Full-Stack & DevOps Engineer

Apr 2026 – Jun 2026

Threadpath Collective (threadpathcollective.ca)

Remote

- Architected end-to-end **full-stack web applications** using React and Node.js microservices, engineering **scalable API integrations** and state management across complex user workflows.
- Developed serverless backend logic using Firebase Cloud Functions, implementing robust **payload validation**, error-handling middleware, and optimized **database queries**. **CI/CD build pipelines** for frontend compilation and backend serverless endpoints.

- Diagnosed and resolved domain-level CDN caching discrepancies across custom endpoints to guarantee **zero-downtime re-leases**.

Lead Developer (Technical Trainee)

Rajiv Gandhi Centre for Biotechnology – IAN-2025 (ian2025.in)

June 2025 – Nov 2025

Thiruvananthapuram

- Built and deployed a Progressive Web App (PWA) managing on-ground logistics for an international neuroscience conference, achieving **100% uptime** during peak volume.
- Automated event workflows: processed **1.2M reads** on Firestore and engineered a digital authentication system executing 2,275+ food coupon scans with zero errors.
- Maintained secure **access controls** (IAN/2025/J21) to isolate sensitive logistical workflows and conference parameters during live execution.
- Developed Python **automation scripts** and data integration routines following strict OOP principles throughout the codebase.

Product Engineering Intern

Lascade LLP – Travel Animator (travelanimator.com)

Jan 2025 – Aug 2025

Kochi

- Spearheaded UI design enhancements and full-stack performance optimizations for **Travel Animator** (travelanimator.com), delivering highly responsive, interactive UI components and intuitive user flows.
- Engineered asset loading pipelines and lightweight animation workflows, significantly reducing **page latency** and optimizing rendering efficiency across low-bandwidth environments.
- Managed Git branching workflows, peer pull-request reviews, and collaborated on **CI/CD pipelines** for automated application builds and staging deployments.

AI/ML Intern

ICT Academy of Kerala

Jul 2024 – Oct 2024

Thiruvananthapuram

- Engineered Python data preprocessing pipelines that boosted image classification accuracy by **10% gain** using advanced data augmentation techniques.
- Fine-tuned facial recognition models via supervised learning, achieving a **5% reduction** in false positive detections.

App Developer & Technical Mentor

Sarvodayam VHSS Arayampadam

Oct 2023 – Mar 2024

Thrissur

- Co-developed the **Ente Naddu** mobile application backed by the Government of Kerala for school students over a 6-month development cycle.
- Mentored student developers across requirement gathering, system architecture, feature implementation, and **code optimization**.

Full Stack Developer Intern

ICT Academy of Kerala

May 2023 – Jul 2023

Thiruvananthapuram

- Architected a responsive MERN stack application, executing full deployment transition from local environments to **cloud infrastructure**.
- Integrated Firebase real-time database listener structures for multi-user data synchronization and **high availability**.

Technical Projects

Agentic Prompt-Ops & System Specs Engine

github.com/AdityaManojA/prompt-ops-blueprints

- Engineered structured **agentic blueprints** and technical specifications to guide autonomous coding agents (Cursor, Windsurf, Claude Code) in generating production-grade architectures.
- Boosted **framework efficiency** across 8+ tech stacks (Next.js, Supabase, Cloudflare Workers, SvelteKit) by enabling LLMs to parse system constraints and interface with Model Context Protocols (MCPs).
- Standardized developer **agentic workflows** by replacing ambiguous natural language instructions with deterministic implementation plans and multi-agent execution graphs.

Education

B.Tech, Computer Science & Engineering

Jyothi Engineering College

2021 – 2025

Thrissur

Achievements & Leadership

- Appointed by Rajiv Gandhi Centre for Biotechnology to lead core digital infrastructure, supporting 2,200+ secure interactions with **100% uptime**.
- Awarded official **Certificate of Appreciation** by Jyothi Engineering College / Govt. of Kerala for technical leadership and mentorship.
- Recognized for **technical excellence** and lab contribution under Dr. Jackson James at the Rajiv Gandhi Centre for Biotechnology.